

Touch-Based Robot-Base Localization with Particle Filtering

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Abstract

This project implements particle filtering to estimate the planar base pose of a seven-degree-of-freedom manipulator from touch observations. Given known obstacle geometry and the joint angles at contact, each particle predicts a probe position whose surface-distance residual determines its likelihood. I implemented the observation model, resampling, perturbation, and an optional online extension that commands the simulated robot to collect contacts. The online strategy combines randomized directional sweeps, contact backoff, and stuck detection. A fresh offline evaluation covers 18 configurations and five trials per configuration. The lowest observed mean combined pose-error score is 0.3369 with 2,500 particles, likelihood width 0.01, and perturbation scale 0.1. Results show interacting effects of likelihood width and perturbation; more particles do not improve every setting. This report separates measured offline results from online implementation behavior and documents limitations of the original metric.

1 Problem and Contribution

The robot base has an unknown but fixed pose $s = (x, y, \theta)$, where x and y describe ground-plane translation and θ is rotation about the vertical axis. A rigid probe with a spherical tip is attached to the manipulator. When the tip touches an obstacle, the seven joint angles form an observation $q \in \mathbb{R}^7$. The task is to recover the base pose from these contact configurations rather than direct pose measurements.

The course framework supplies a PyBullet simulation, forward-kinematics solver, prerecorded observations, and interfaces for Cartesian motion and contact detection. I implemented surface-distance calculations, particle likelihoods, offline and online filters, and the online observation-collection strategy. The filtering structure and likelihood model follow the assignment; the sweep strategy is my implementation choice within its optional online task. This is a course-project implementation and evaluation, not a claim of a new filtering algorithm.

2 Observation Model

The environment contains five cylindrical obstacles with known planar centers c_j and radii r_j . The probe-tip radius is $r_p = 0.02$. Forward kinematics maps the observed joint angles to the probe-tip position $u(q)$ in the base frame. Particle $s_i = (x_i, y_i, \theta_i)$ predicts its world position as

$$p_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} \cos \theta_i & -\sin \theta_i \\ \sin \theta_i & \cos \theta_i \end{bmatrix} u(q). \quad (1)$$

For each obstacle, the signed surface-distance residual is

$$d_{ij} = \|p_i - c_j\|_2 - (r_p + r_j). \quad (2)$$

The implementation selects the residual closest to zero in absolute value. Contact yields zero, separation a positive value, and penetration a neg-

ative value. With $j^* = \arg \min_j |d_{ij}|$, normalized weights are

$$w_i = \frac{\exp[-d_{ij^*}^2 / (2\sigma^2)]}{\sum_k \exp[-d_{kj^*}^2 / (2\sigma^2)]}. \quad (3)$$

Here j^* is the selected obstacle for particle k . The Gaussian prefactor cancels during normalization. The code evaluates the density directly and falls back to uniform weights if all likelihoods numerically vanish. This avoids division by zero but loses information from that update; log-domain normalization would provide stronger numerical handling.

3 Sequential Estimation

Particles are initialized uniformly over $[-1, 1] \times [-1, 1] \times [-\pi, \pi]$. For each observation, the filter computes likelihoods and draws N particles with replacement using the normalized weights. It then adds Gaussian perturbations with covariance $\delta^2 I_3$. Resampling favors contact-consistent hypotheses; perturbation maintains local diversity among duplicated particles.

After perturbation, position coordinates are clipped to $[-1, 1]$ and angles wrapped to $[-\pi, \pi]$. Weights are reset uniformly after each update. The offline filter processes the supplied observations once and returns the arithmetic mean of final particle coordinates. No convergence-based early stopping is implemented. Because the pose is static, perturbation serves as particle roughening rather than a model of physical base motion.

4 Online Observation Collection

The optional extension replaces prerecorded observations with contacts collected during simulation. For each request, the controller considers eight evenly spaced planar directions, randomly rotated at the start of each of up to four rounds. Before each attempted direction, it resets the joints to a nominal configuration and commands a Cartesian sweep at speed 0.12 in simulation units.

A detected tip contact records the seven joint angles and triggers 20 reverse-velocity steps before returning the observation. If planar tip displacement stays below 10^{-4} for 40 consecutive steps, the controller abandons that direction. Each sweep permits at most 1,200 steps. Exhausting all attempts raises an explicit failure rather than running indefinitely.

The online filter performs 100 observation updates unless collection fails. The strategy uses supplied simulation controls; it does not plan collision-free return motions or select actions from current particle uncertainty. Backoff separates the tip from contact but does not establish statistical independence between observations. Collection logs track attempts and simulation steps, not localization accuracy or convergence speed.

5 Offline Evaluation

5.1 Protocol and Error Definition

Every run uses the same prerecorded observation file and ground-truth pose $s^* = (-0.3, -0.3, 0.9)$. The assignment supplies 100 contact configurations. Task 5 evaluates $N \in \{1000, 2500\}$, $\sigma \in \{0.01, 0.05, 0.2\}$, and $\delta \in \{0.001, 0.01, 0.1\}$: 18 configurations and 90 runs. Each trial creates a fresh headless simulation and particle initialization with plotting disabled.

For trial t , the reported score is

$$e_t = \|\hat{s}_t - s^*\|_2, \quad \bar{e} = \frac{1}{5} \sum_{t=1}^5 e_t. \quad (4)$$

The standard deviation uses NumPy’s population convention, $\sqrt{\sum_t (e_t - \bar{e})^2 / 5}$. These are final-estimate scores across trials, not averages over iterations. The mean trial error also differs from the error of the mean estimated pose.

The metric combines translation and angle differences without scaling or angular wrapping. It is a combined pose-error score, not a distance in meters, and is retained to report the supplied experiment faithfully.

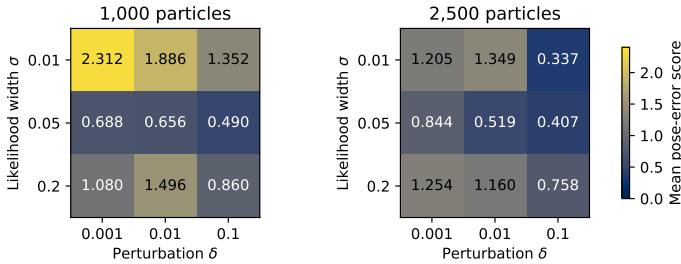


Figure 1: Fresh offline sweep: mean final pose-error score across five trials. Both panels share a color scale; lower is better. Rows vary σ and columns vary δ .

5.2 Results and Interpretation

The lowest observed mean error is 0.3369 at $N = 2500$, $\sigma = 0.01$, $\delta = 0.1$. Its across-trial standard deviation, 0.0716, is also the lowest in this batch. The corresponding mean estimated pose is $(-0.267912, -0.177448, 1.211134)$. This summarizes five final estimates; it is not a separate trial.

For every fixed particle-count and likelihood-width pair, $\delta = 0.1$ has the lowest mean error among the tested perturbation scales. A narrow likelihood is therefore not uniformly harmful: the winning configuration combines the narrowest likelihood with the largest perturbation. In contrast, at $N = 1000$, $\sigma = 0.01$, $\delta = 0.001$, mean error reaches 2.3118. These results support evaluating parameters jointly rather than choosing each in isolation.

Increasing particle count reduces mean error in seven of nine matched (σ, δ) settings but increases it in two. The best 1,000-particle configuration uses $\sigma = 0.05$, $\delta = 0.1$, with mean error 0.4902 and standard deviation 0.1582. Five trials per setting support descriptive comparisons, not statistical superiority or general robustness claims.

Table 1: Complete offline sweep. Each row summarizes five trials. Mean and SD refer to the original combined pose-error score; bold marks the lowest observed mean.

N	σ	δ	Mean	SD
1000	0.01	0.001	2.3118	1.2916
1000	0.01	0.010	1.8859	1.2828
1000	0.01	0.100	1.3524	1.0433
1000	0.05	0.001	0.6882	0.6803
1000	0.05	0.010	0.6563	0.8422
1000	0.05	0.100	0.4902	0.1582
1000	0.20	0.001	1.0803	0.4064
1000	0.20	0.010	1.4963	0.4320
1000	0.20	0.100	0.8597	0.4727
2500	0.01	0.001	1.2051	0.4254
2500	0.01	0.010	1.3489	1.3921
2500	0.01	0.100	0.3369	0.0716
2500	0.05	0.001	0.8437	0.5826
2500	0.05	0.010	0.5189	0.6379
2500	0.05	0.100	0.4070	0.1540
2500	0.20	0.001	1.2541	0.2850
2500	0.20	0.010	1.1601	0.5152
2500	0.20	0.100	0.7583	0.3531

6 Limitations and Reproducibility

Fresh evaluation. Tables and heatmaps use the fresh rerun supplied for this revision, superseding the original numerical summaries. No fixed seed was recorded; exact reproduction is not guaranteed. The accompanying source bundle retains the full-precision summary.

Orientation handling. Arithmetic averaging can misrepresent angles straddling $-\pi$ and π . A future version should use a circular mean and report translation error separately from wrapped angular error. These changes have not been applied retroactively.

Scope of evidence. The sweep uses one environment, pose, and observation sequence. It measures final error, not runtime or convergence speed. No repeated-run online comparison is available: offline scores do not quantify the benefit of sweeps. The controller uses simulation resets and does not demonstrate hardware deployment.

Execution settings. Run `python main.py --task 5` for the fixed sweep. Task 3 collects observations online with a 1/500-second timestep. The driver defaults to 1,000 particles, overriding the online function’s 3,000-particle default; pass `--num_particles 3000` to select 3,000.

7 Conclusion

The implementation combines geometric contact likelihoods, particle filtering, and bounded online probing. Across 90 offline runs, likelihood width and perturbation interact, and additional particles do not improve every setting. The project demonstrates estimation and observation-collection engineering with explicit numerical and simulation limitations.

Project Materials

[1] Rice University, COMP 462/562. *Project 3: State Estimation*. Course assignment and simulation framework. <https://github.com/robotpilot/COMP462/tree/main/project3>.

[2] Zarek Lu. *COMP 562 Project 3*. Original course report, April 2026; supplied `alg.py`, `main.py`, and fresh Task 5 summary. Tables and figures here are regenerated from that summary.